D 11376 **(Pages : 3)** **Name.....**

Reg. No.....

THIRD SEMESTER M.Sc. DEGREE (SUPPLEMENTARY) EXAMINATION NOVEMBER 2021

(CUCSS)

Mathematics

MT 3C 14—FUNCTIONAL ANALYSIS

(2016 to 2018 Admissions)

Time : Three Hours Maximum : 36 Weightage

Part A

Answer **all** *the questions.*
Each question carries weightage 1.

1. Show that the metric space l^∞ is not separable.
2. Show that for $p \geq 1$, the sequence space $l^p \subset l^\infty$.
3. Show by an example that a linear map on a linear space X may be continuous with respect to some norm on X, but discontinuous with respect to another norm on X.
4. State the Hahn-Banach extension theorem.
5. Is c_{00}, the space of all scalar sequences having only finitely many non-zero terms as a subspace of l^∞, a closed set ? Justify your answer.
6. State the Uniform boundedness principle.
7. State the parallelogram law in inner product spaces.
8. What is known as the Minkowski space ?
9. State the Minkowski's inequality for sequences.
10. Define an essentially bounded function on a set.
11. Show that the metric space $L^p[a, b]$ is separable for $1 \leq p < \infty$.
12. Show that the function $\| \ \|$, is uniformly continuous on a normed space X.

Turn over

13. Show that the hypothesis of completeness on the domain cannot be dropped in the Uniform boundedness principle.

14. Let $\langle , \rangle$ be an inner product on a linear space X. If $\| x_n - x \| \to 0$ and $\| y_n - y \| \to 0$, then, show that $\langle x_n, y_n \rangle \to \langle x, y \rangle$.

(14 × 1 = 14 weightage)

Part B

Answer any **seven** *questions.*
Each question carries weightage 2.

15. Give an example to show that the open mapping theorem may not hold true if the normed spaces X is not a Banach space even when Y is a Banach spaces.

16. Show that a linear map F from a normed space X to a normed space Y is a homeomorphism if there are $\alpha, \beta > 0$ such that $\beta \| x \| \leq \| F(x) \| \leq \alpha \| x \|$ for all $x \in X$.

17. Let X and Y be normed spaces. If Z is a closed subspace of X, then show that the quotient map Q from X to X/Z is continuous and open.

18. Let X be a normed space, Y be a closed subspace of X and $Y \neq X$. Let r be a real number such that $0 < r < 1$. Show that there exists some $x_r \in X$ such that $\| x_r \| = 1$ and $r \leq dis(x_r, Y) \leq 1$.

19. State and prove the Polarization identity in an inner product space.

20. Let X be an inner product spaces. Let $\{x_1, x_2, ..., x_n\}$ be an orthogonal set in X, then show that $\| x_1 + x_2 + ... + x_n \|^2 = \| x_1 \|^2 + \| x_2 \|^2 + ... + \| x_n \|^2$.

21. Give an example to show that in a non-complete metric space, the intersection of a denumerable number of dense open subsets need not be dense.

22. Let X be a normed space and E be a convex subset of X. Show that the closure $\overline{E}$ of E is convex.

23. When do you say that a linear functional is positive ? Give an example for a positive linear functional.

24. Let X be a normed space over K, and f be a non-zero linear functional on X. If E is an open subset of X, then show that $f(E)$ is an open subset of K.

(7 × 2 = 14 weightage)

Part C

Answer any **two** *questions.*
Each question carries weightage 4.

25. Let X be a metric space. Then show that :

    (i) The intersection of a finite number of dense open subsets of X is dense in X.

    (ii) If X is complete, then the intersection of a countable number of dense open subsets of X is dense in X.

26. State and prove the open mapping theorem.

27. State and prove the Gram-Schmidt orthoganalization theorem.

28. Let X and Y are normed spaces and $X \neq 0$. Then show that :

    (a) $BL(X, Y)$, the set of all bounded linear maps from X to Y is a Banach space in the operator norm if and only if Y is a Banach space.

    (b) The dual X' of every normed space X is a Banach space.

(2 × 4 = 8 weightage)